

KHALID AHMAD

Lokoja, Nigeria.

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Research Interests

Applied Machine Learning, Explainable and Trustworthy AI, Fairness and Bias in Medical AI

Education

Federal University, Lokoja

Sep. 2018 – Oct. 2023

B.Sc. Computer Science, **First Class Honours**, CGPA: 4.75/5.0 (95%)

Lokoja, Nigeria

Class Rank: 2/159 (Top 1% in the Department, Top 1% Overall)

Research Experience

Federal University, Lokoja

Oct 2023 – Present

Research Assistant

Lokoja, Nigeria — Hybrid

- Built a road anomaly detection system using a DeepConvAutoencoder and Vision Transformer for real-time defect detection.
- Developed a predictive model using academic and non-academic factors to estimate first-year GPA for improved advising.

Wale University

Feb 2025 – Jun 2025

Research Intern

Lagos, Nigeria — Remote

- Conducted systematic literature reviews on AI applications in healthcare and analyzed existing machine learning models for the early detection of sickle cell disease, with evaluation based on performance metrics and clinical relevance.

Publications

- **Khalid, H.A.**, Fati, O.O, Abdulhakeem A.O, & Aifuobhokhan, J. (Manuscript in Progress). An Interpretable Boosting Framework for Automated Classification of Pattern Electroretinogram Signals with Clinical Metadata.
- **Khalid, H.A.**, Temitope, O.K, Fati O.O, & Olutomilayo, O.P. (Under Review). A Lightweight Explainable AI Model for Liver Fibrosis Classification with Dilated Attention.
- Ashiru, A.Z., Mustapha, A., Muhammad, B.A., Aliyu, H.I., & **Khalid, H.A.** (Under review). Discovery of Breast Cancer Biomarkers Using Machine Learning and Systems Biology. *Discover Biotechnology*
- Aifuobhokhan, J., **Khalid, H.A.**, Ekechi, C.C., Olanrewaju, A.O., Afuadajo, E., Bowale, D.A., & Inioluwa, O.M. (2025). Predicting Thyroid Cancer Recurrence Using Machine Learning *International Journal of Advances in Artificial Intelligence and Machine Learning* [DOI].
- Aderemi, I.A., Kehinde, T.O., Ugochukwu, D.O., **Khalid, H.A.**, Adjei, K.Y., & Chijioke, C.E. (2025). Beyond the Black Box: A Systematic Review of Explainable AI for Transparent and Trustworthy Water Quality Monitoring. *IEEE Sensors Reviews* [DOI].
- **Khalid, H.A.**, & Fati, O.O. (2025). A Real-Time Road Anomaly Detection via Masked Autoencoder and Enhanced Vision Transformer. *International Journal of Innovative Research in Engineering & Management* [DOI].

Conferences & Presentations

- **Khalid, H.A.**, Fati, O.O., Faruk, A., & Ismail, A (2025). Machine Learning Models for First-Year GPA Prediction in Nigerian Tertiary Institutions. *International Conference on Artificial Intelligence and Robotics (ICAIR)* [Link].
- **Khalid, H.A.**. Exploring Technology Hotspots: The Impact of Mathematics on Computer Science. *National Association of Computing Students (NACOS) Workshop, FUL Chapter*, Aug 2022

Academic & Professional Service

Journal of Electrical Systems and Information Technology

2025

Reviewer

- Reviewing submitted manuscripts in machine learning and applied artificial intelligence.

Deep Learning Indaba

Jul 2025 – Aug 2025

Judge

Kigali, Rwanda — Hybrid

- Served as a judge for the Deep Learning Indaba 2025 AI Community Challenge in Kigali, contributing to the review of more than 160 teams from over 20 countries and supporting efforts to advance innovative AI research in Africa.

Teaching Experience

Digital Onboarders

Jan 2024 – Nov 2024

CS Mentor & AI Workshop Facilitator

Lafia, Nigeria — Hybrid

- Mentored 78 students in Python, web development, and AI fundamentals through hands-on sessions, workshops, and national events, including the Miva AI Conference.

Federal University Lokoja

Mar 2019 – Jul 2023

Academic Tutor

Lokoja, Nigeria — Onsite

- Tutored 2,500+ students in core STEM and computer science courses, improving student performance through personalized instruction and collaborative faculty support.

Industrial Experience

Omdena

Feb 2025 – May 2025

Machine Learning Engineer

New York, USA — Remote

- Collaborated with 10 engineers on a Smart Farming project for 10,000+ farmers in Kano State, cleaning 148,666 soil moisture records and training a Random Forest model that improved plant health detection accuracy by 9% to 99% precision.

Engineeum

Jan 2025 – Mar 2025

Machine Learning Instructor

Lagos, Nigeria — Remote

- Taught data analysis and machine learning in Python using NumPy, Pandas, Matplotlib, and Seaborn, covering supervised/unsupervised learning and time series, while designing course materials, guiding real-world projects, and providing feedback to enhance student outcomes.

Saint Louis University

Jun 2023 – Jul 2023

Data Analyst

Missouri, USA — Remote

- Analyzed 10 digital marketing campaigns for Finance and Marketing teams, built interactive dashboards in Python, SQL, and Power BI, and reported ad spend and engagement metrics, enabling targeted optimizations that improved campaign performance.

Nigerian Midstream & Downstream Petroleum Regulatory Authority

Mar 2022 – Aug 2022

Data Scientist Intern

Lokoja, Nigeria — Onsite

- Analyzed 155 petrol samples from 39 Lokoja stations, built ML models (Logistic Regression, Decision Trees, Random Forest, SVM) achieving 75% accuracy, and integrated predictive analytics to reduce manual inspections by 30%.

Technical Skills

Programming & Tools: Python, C++, JavaScript, HTML, CSS, SQL, Git, Docker, VS Code, Jupyter, Azure

Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, OpenCV, NumPy, Pandas, NLTK, Matplotlib, Hugging Face

Other Skills: LaTeX, Microsoft Word, Google Docs, Problem solving, Critical thinking, Team collaboration, Research and documentation, Communication, Time management, Attention to detail, Adaptability

Honors & Awards

- Wale Lab Research Internship – 2025: Fully funded AI healthcare research internship.
- Departmental and Faculty Fee Waiver – 2023: Awarded for top academic performance and engagement.
- Ayomide Ajiboye Scholar (USA) – 2023: Scholarship and mentorship for academic excellence.
- Student Tutor of the Year – 2023: Recognized for outstanding peer tutoring.
- Department Contribution Award – 2022: Honored for impactful research and academic service.
- NMDPRA Recommendation – 2022: Commended for exceptional work with Nigeria's petroleum authority.
- Dean's Honor Award – 2020–2023: Multiple-year award for sustained academic excellence.
- Kogi State Bursary Scholar – 2020: Financial award for academic merit and need.

Professional Affiliations

- Member, Black in AI (BAI)
- Member, Data Science Nigeria (DSN)
- Member, National Association of Computing Students (NACOS)
- Member, Nigeria Computer Society (NCS)
- Core Member, Google Developer Student Clubs (GDSC)

Referees

- Available upon request